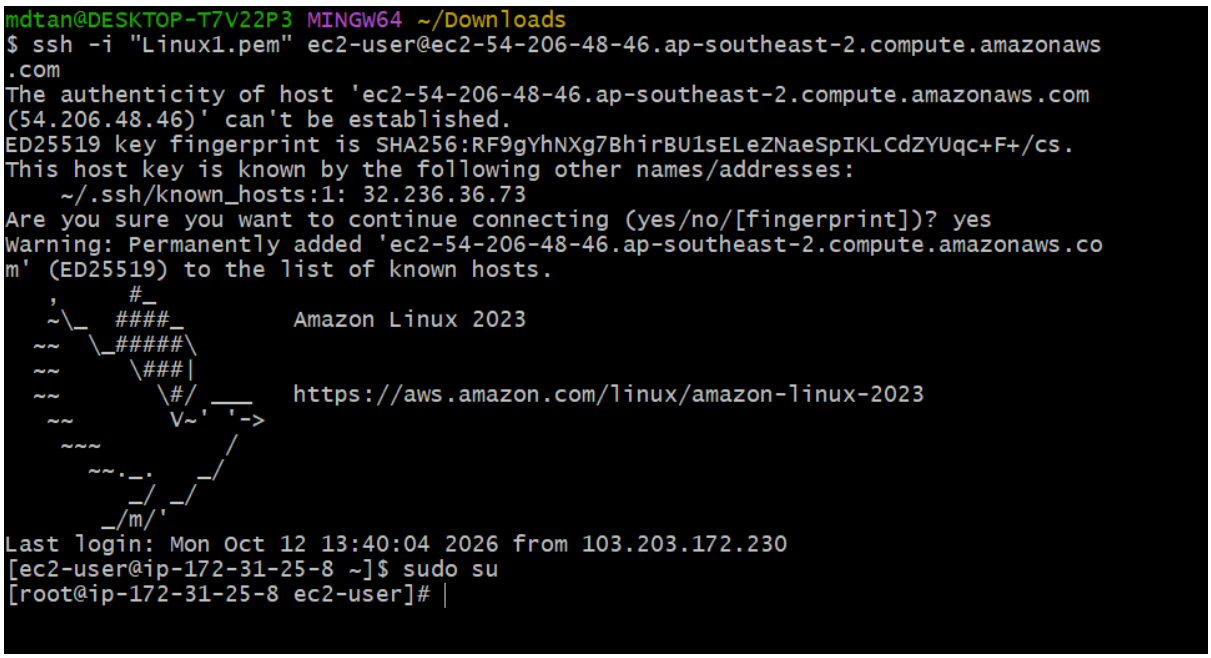
Bash-Scripting assignment-01.

Notes: ***Create a Jira ticket for this assignment, work on the tasks listed below, and submit the task completion document by EOD with clear steps and required evidence (outputs/screenshots)***

1. Watch the video and write down notes.
2. Execute all scripts shown in video and capture the screenshots.
3. Execute all the scripts share in #devops-cloud-documents-14 channel.

Step-by-Step Execution Sequence

1. **Server Connection & Root Access**
    - **Action:** Connects to an Amazon Linux 2023 EC2 instance via SSH using the private key file Linux1.pem, and elevates privileges to the root user using sudo su.
    - **Verification:** The terminal prompt changes to root@ip-... indicating successful administrative access.



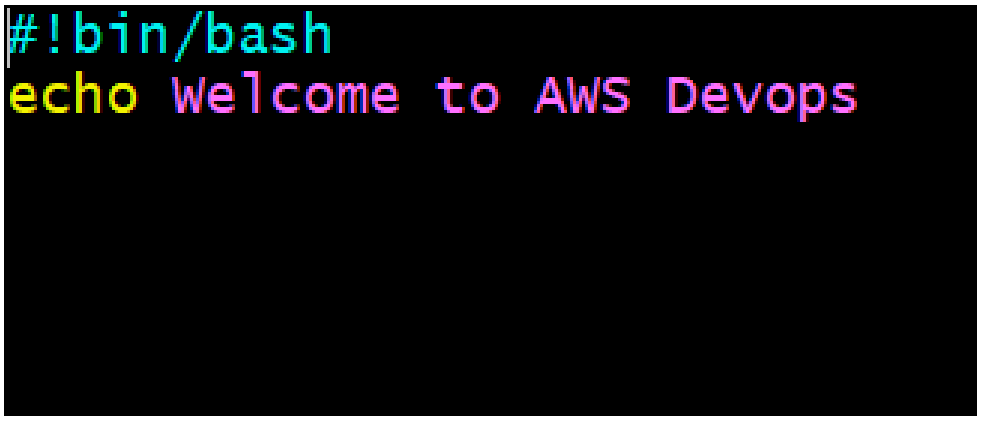
Shell Identification & Verification

- **Action:** Checks the current shell environment by running echo $0 (which returns bash) and inspects available system shells by reading /etc/shells.
- **Verification:** Lists valid shell paths like /bin/bash, /bin/sh, and /bin/tcsh

```
[root@ip-172-31-25-8 ec2-user]# echo $0
bash
[root@ip-172-31-25-8 ec2-user]# cat /etc/shells
/bin/sh
/bin/bash
/usr/bin/sh
/usr/bin/bash
/bin/csh
/bin/tcsh
/usr/bin/csh
/usr/bin/tcsh
[root@ip-172-31-25-8 ec2-user]#
```

Writing Basic Scripts & Comments

- **Action:** Creates script files starting with the proper shebang line (#!/bin/bash) followed by comments for documentation (purpose, date) and basic text output using echo.
- **Verification:** Assigns variables (e.g., a='hello world') and prints them successfully.



Managing File Permissions (chmod)

- **Action:** Creates custom scripts (e.g., dell.bash), checks file permissions using ll, and encounters a "Permission denied" error when trying to run ./admin.bash directly.
- **Verification:** Fixes permission issues by granting executable rights using/

Executing System Information & Utility Commands

- **Action:** Runs scripts (commands.bash) that output environmental details, including the current directory (pwd), file listings (ls), user identity (whoami), system date, and calendar (cal).

```
cat: welcome.bash: No such file or directory
[root@ip-172-31-25-8 scripts]# cat welcome to AWS Devops
#!/bin/bash
# purpose:Testing script format
# Date 15 october 2026
# done successfully
a='hello world'

echo $a
```

```
root@ip-172-31-25-8 scripts]# vi dell.bash
root@ip-172-31-25-8 scripts]# ll
otal 8
rw-r--r--. 1 root root 171 Oct 15 10:58 dell.bash
rwxrwxrwx. 1 root root 157 Oct 15 10:41 welcome
root@ip-172-31-25-8 scripts]# chmod 755 dell.bash
root@ip-172-31-25-8 scripts]# ./dell.bash
elcome to dell laptops
root@ip-172-31-25-8 scripts]#
```

```
Welcome to dell laptops
[root@ip-172-31-25-8 scripts]# vi commands.bash
[root@ip-172-31-25-8 scripts]# chmod 755 commands.bash
[root@ip-172-31-25-8 scripts]# ./commands.bash
/tanveer/scripts
commands.bash  dell.bash  welcome
root
Thu Oct 15 11:09:11 UTC 2026
    October 2026
Su Mo Tu We Th Fr Sa
             1  2  3
 4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

[root@ip-172-31-25-8 scripts]#
```

System Administration & Performance Monitoring

- **Action:** Executes administrative scripts (admin.bash) that leverage system diagnostics tools like top (process activity) and disk/memory utilization metrics (df -h, free).
- **Verification:** Displays real-time CPU usage, memory stats, and system load averages.

```
[root@ip-172-31-25-8 scripts]# ./admin.bash
bash: ./admin.bash: Permission denied
[root@ip-172-31-25-8 scripts]# chmod 755 admin.bash
[root@ip-172-31-25-8 scripts]# ./admin.bash
top - 11:36:14 up  2:32,  2 users,  load average: 0.01, 0.01, 0.00
Tasks: 110 total,   1 running, 108 sleeping,   1 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    913.2 total,    537.3 free,    164.2 used,    211.7 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    617.3 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  173924  17844  11236 S   0.0   1.9   0:01.44 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.04 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.11 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.03 migration/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     21 root      rt   0       0      0      0 S   0.0   0.0   0:00.06 migration/1
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.04 ksoftirqd/1
     24 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     28 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
     29 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     30 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     31 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     32 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     34 root      20   0       0      0      0 S   0.0   0.0   0:00.00 khungtaskd
     35 root      20   0       0      0      0 S   0.0   0.0   0:00.00 oom_reaper
     37 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-writeback
```

```
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           457M     0  457M   0% /dev/shm
tmpfs           183M  440K  183M   1% /run
efivarfs        128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1  8.0G  1.6G  6.4G  20% /
tmpfs           457M     0  457M   0% /tmp
/dev/nvme0n1p128  10M  1.3M  8.7M  13% /boot/efi
tmpfs            92M     0   92M   0% /run/user/1000
               total        used        free      shared  buff/cache   available
Mem:           913Mi       164Mi       537Mi       0.0Ki       211Mi       617Mi
Swap:             0B          0B          0B
 11:36:39 up  2:33,  2 users,  load average: 0.01, 0.01, 0.00
Linux 6.18.51-120.163.amzn2023.x86_64 (ip-172-31-25-8.ap-southeast-2.compute.internal)  10/15/26        _x86_64_        (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
          0.10    0.00    0.06    0.02    0.11   99.72

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1          1.44        23.74        11.59         0.00     217995     106408          0
```

```
run variables tasks

Thu Oct 15 11:47:08 UTC 2026
Thu Oct 15 11:47:08 UTC 2026
                                2026

       January               February                March
Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa
             1  2  3    1  2  3  4  5  6  7    1  2  3  4  5  6  7
 4  5  6  7  8  9 10    8  9 10 11 12 13 14    8  9 10 11 12 13 14
11 12 13 14 15 16 17   15 16 17 18 19 20 21   15 16 17 18 19 20 21
18 19 20 21 22 23 24   22 23 24 25 26 27 28   22 23 24 25 26 27 28
25 26 27 28 29 30 31                          29 30 31

        April                  May                   June
Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa
          1  2  3  4                   1  2       1  2  3  4  5  6
 5  6  7  8  9 10 11    3  4  5  6  7  8  9    7  8  9 10 11 12 13
12 13 14 15 16 17 18   10 11 12 13 14 15 16   14 15 16 17 18 19 20
19 20 21 22 23 24 25   17 18 19 20 21 22 23   21 22 23 24 25 26 27
26 27 28 29 30         24 25 26 27 28 29 30   28 29 30
                       31

        July                 August                September
Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa
          1  2  3  4                      1          1  2  3  4  5
 5  6  7  8  9 10 11    2  3  4  5  6  7  8    6  7  8  9 10 11 12
12 13 14 15 16 17 18    9 10 11 12 13 14 15   13 14 15 16 17 18 19
19 20 21 22 23 24 25   16 17 18 19 20 21 22   20 21 22 23 24 25 26
26 27 28 29 30 31      23 24 25 26 27 28 29   27 28 29 30
                       30 31

       October              November              December
Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa   Su Mo Tu We Th Fr Sa
             1  2  3    1  2  3  4  5  6  7          1  2  3  4  5
 4  5  6  7  8  9 10    8  9 10 11 12 13 14    6  7  8  9 10 11 12
11 12 13 14 15 16 17   15 16 17 18 19 20 21   13 14 15 16 17 18 19
18 19 20 21 22 23 24   22 23 24 25 26 27 28   20 21 22 23 24 25 26
25 26 27 28 29 30 31   29 30                 27 28 29 30 31

[root@ip-172-31-25-8 scripts]# |
```

Handling User Input (read command)

- **Action:** Creates interactive scripts (input.bash) that prompt the user for input using the read command.
- **Verification:** Captures user responses dynamically and prints personalized output strings.

```
[root@ip-172-31-25-8 scripts]# vi input.bash
[root@ip-172-31-25-8 scripts]# chmod 755 input.bash
[root@ip-172-31-25-8 scripts]# ./input.bash
hello, my name is mohammed tanveer ahmed

what is your name?
ahmed

hello ahmed

my Institute name is techie horizon

And yours?
dataminds

[root@ip-172-31-25-8 scripts]# |
```